How Redundant Are Transformer Layers?

Attention vs MLP Redundancy Across Decoder-Only and Seq2Seq Models

Cochiorca Oana Maria
Lungu Laura-Vanesa
Negoiță-Crețu Raluca-Marina

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Introduction

Motivation & Research Questions

Introduction



LLMs are powerful, but expensive

Large language models achieve strong performance, but their inference cost limits real-world deployment.



Not all Transformer layers matter equally

"What Matters in Transformers?" shows that large decoder-only models contain redundant attention layers that can be dropped with minimal performance degradation.



Does attention redundancy generalize?

We ask whether attention-layer redundancy persists across smaller models, multilingual fine-tuned models, and encoder–decoder architectures.

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Related Work

Related Work



Quantization (industry standard)

Reduces memory and inference cost by using low-precision weights, but keeps the full architecture unchanged.



Pruning and Distillation

Pruning removes parameters or layers, and distillation compresses models but requires costly retraining.



Architectural Redundancy in Transformers

Recent work shows many Transformer layers - especially deeper attention layers - contribute little to performance and can be removed safely.

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Methodology Overview

Hypotheses

Goal

Test whether *attention-layer redundancy* generalizes
across **scale, language, and architecture**

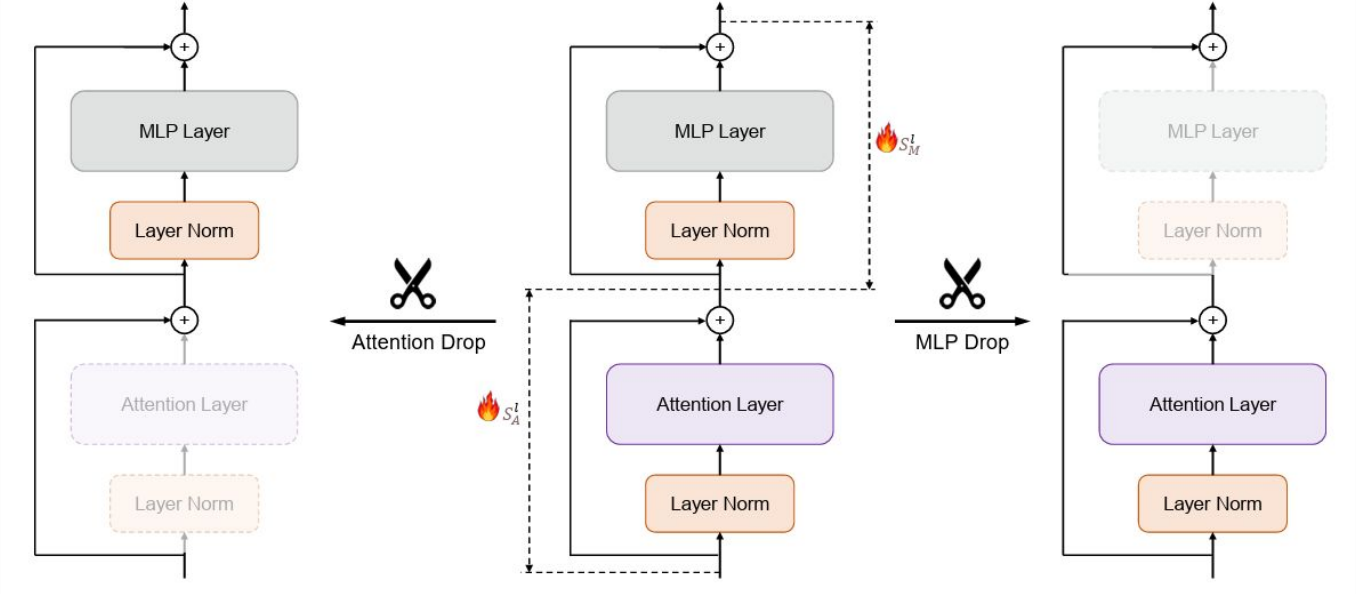
Models evaluated

- **RoLLaMA2-7B** — decoder-only, fine-tuned for Romanian
- **TinyLlama-1.1B** — small decoder-only baseline
- **mT5-small** — encoder–decoder (seq2seq)

Approach

- Measure **layer importance** using similarity-based calibration
- Apply **structured layer ablations (Attention / MLP / Joint drop)**
- Evaluate impact on downstream benchmarks

Measuring Layer Importance



Because Transformer layers use residual connections, we measure layer importance by comparing the input and output of each sublayer (after the residual addition), so layers that barely change the representation are identified as redundant.

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Experiments & Results

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RoLLaMA2-7B-Base

Decoder-only model

RoLLaMA2-7B-Base - Experimental Setup

Calibration Dataset

Calibration on Romanian Subset from wikimedia/wikipedia corpus (256 samples × 2048 tokens)

Evaluation

Zero-shot evaluation on benchmarks:

- ARC-Easy / ARC-Challenge (reasoning)
- Hella Swag (commonsense)
- WinoGrand (coreference)

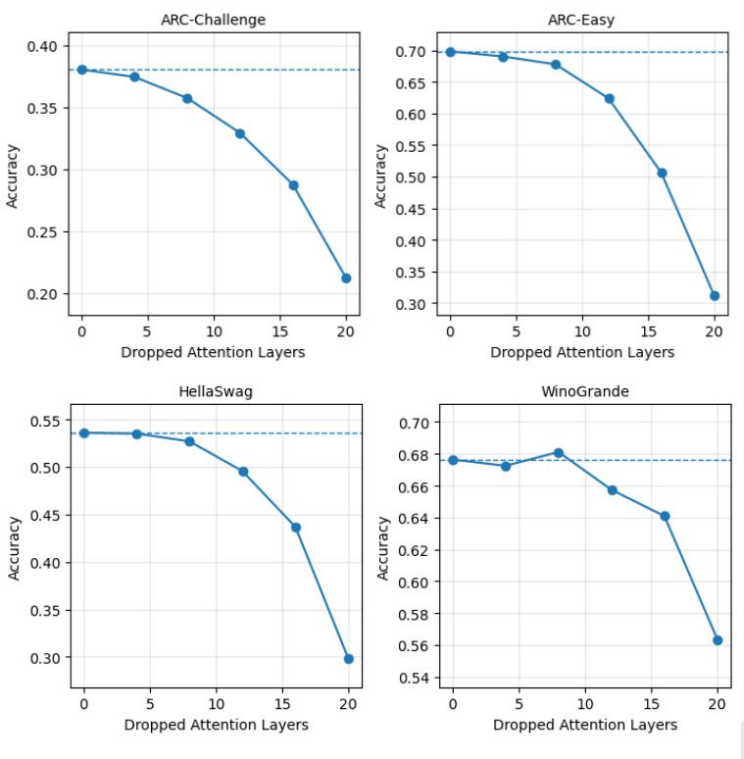
Ablation Configurations

- **Attention Drop** (removed 8, 12, 16, and 20 most redundant attention layers.)
- **MLP Drop** (removed 4, 8, and 12 most redundant MLP layers)
- **Joint Drop** (removed 12, 16, 24 and 32 most redundant attention and MLP layers)

Attention Layer Drop Results

- Remove up to **25% of attention layers**
- **>90% of baseline performance preserved**
- Degradation is gradual, not catastrophic
- Strong robustness on HellaSwag & WinoGrande

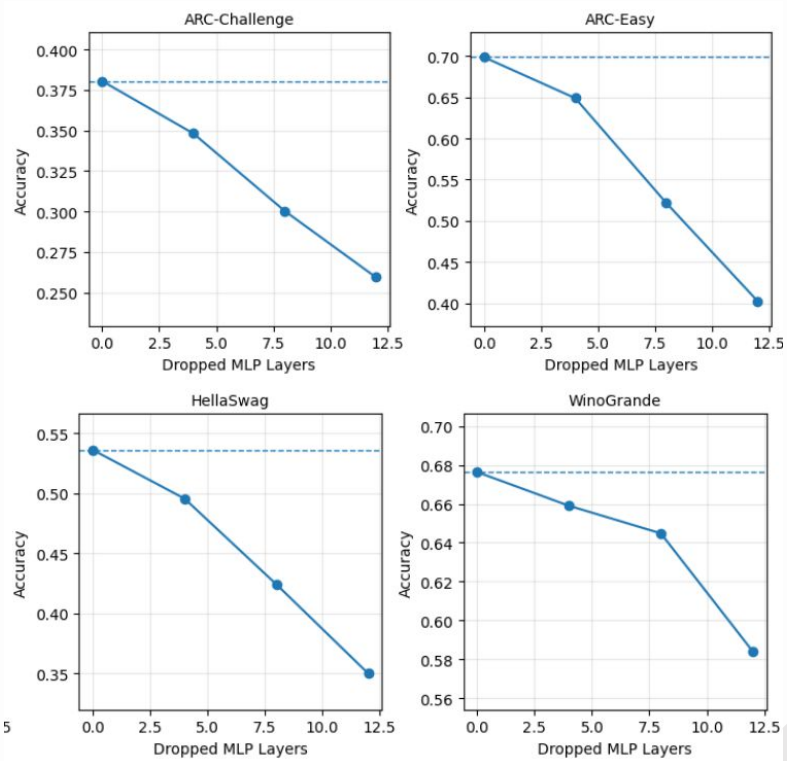
=> **Attention layers show high redundancy**



MLP Layer Drop Results

- Performance drops much faster
- Even small MLP removal hurts accuracy
- Confirms MLPs carry most model capacity

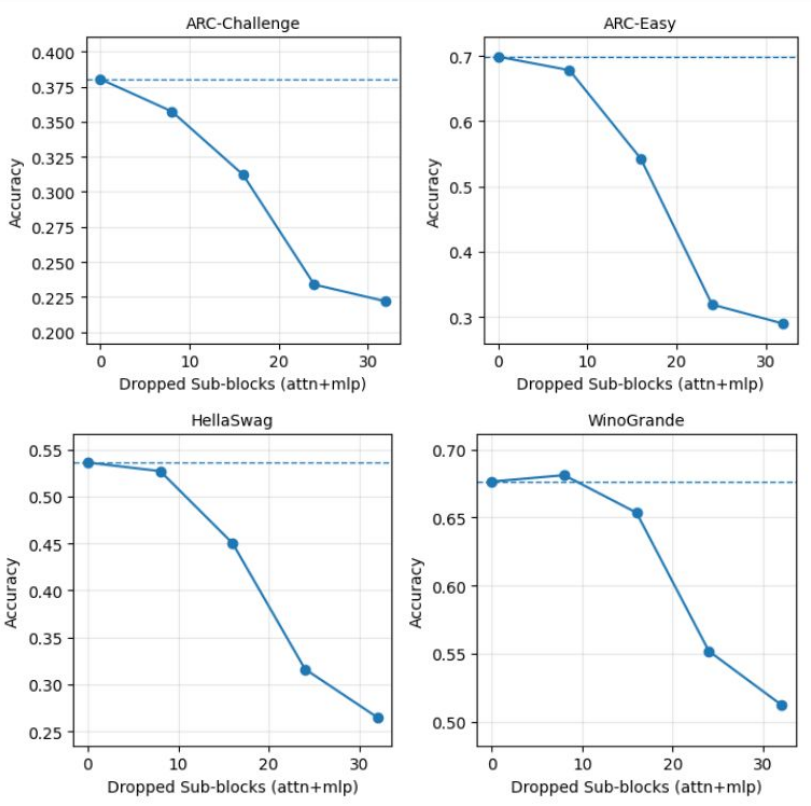
=> **MLP layers are far less redundant than attention**



Joint Layer Drop Results

- Combines attention + MLP removal
- Enables stronger compression
- Works up to moderate pruning
- Sharp accuracy collapse beyond some point

=> **Aggressive compression requires careful limits**



Key findings

- Attention layers are consistently more redundant than MLP layers
- Previous findings hold for a **fine-tuned, non-English model**
- Confirms redundancy is **not language-specific**
- Supports structured pruning as a viable efficiency strategy

Method	ARC-C	ARC-E	HellaSwag	WinoGrande
Baseline	0.381	0.699	0.536	0.676
Attn-4	0.375	0.691	0.535	0.673
Attn-8	0.358	0.678	0.527	0.681
Attn-12	0.329	0.624	0.496	0.658
Attn-16	0.288	0.506	0.437	0.641
Attn-20	0.213	0.311	0.298	0.564
MLP-4	0.348	0.649	0.495	0.659
MLP-8	0.300	0.522	0.424	0.645
MLP-12	0.259	0.402	0.350	0.584
Joint-8	0.358	0.678	0.527	0.681
Joint-16	0.312	0.543	0.450	0.654
Joint-24	0.234	0.319	0.316	0.552
Joint-32	0.222	0.290	0.265	0.512

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TinyLlama-1.1B

Decoder-only model

Experimental Setup

TinyLlama-1.1B

- 22-layer decoder-only Transformer
- Used to test whether attention redundancy persists at smaller scale

Calibration Dataset

Four diverse corpora (256 samples × 2048 tokens each):

- wiki_ro & wiki_en → clean encyclopedic text
- mC4_ro & mC4_en → diverse web tex

Pruning Strategies & Ratios

Attention Drop: 10% / 25% / 50% (2 / 6 / 11 layers)
MLP Drop: 10% / 25% / 50% (2 / 6 / 11 layers)
Joint Drop

- Rank attention + MLP together
- Remove 4 / 11 / 22 modules total

Evaluation

Zero-shot benchmarks:

- BoolQ (yes/no questions)
- HellaSwag (commonsense reasoning)
- PIQA (physical commonsense)
- WinoGrande (coreference resolution)

Plus: Perplexity on Romanian mC4 (language modeling quality)

Calibration Data & Insights



Insights

MLP layers are more important

Attention median score: 0.044
MLP median score: 0.124 → MLPs are ~3× more critical than attention

Attention redundancy is intrinsic

Attention rankings highly stable across all corpora: Spearman ρ = 0.80–0.97 (cross-lingual & cross-domain)

MLP importance varies by domain

MLP layer rankings less stable than attention. Importance depends on calibration data.

Deeper attention layers are most redundant

Calibration identifies deepest attention layers as least important → safe to remove first.

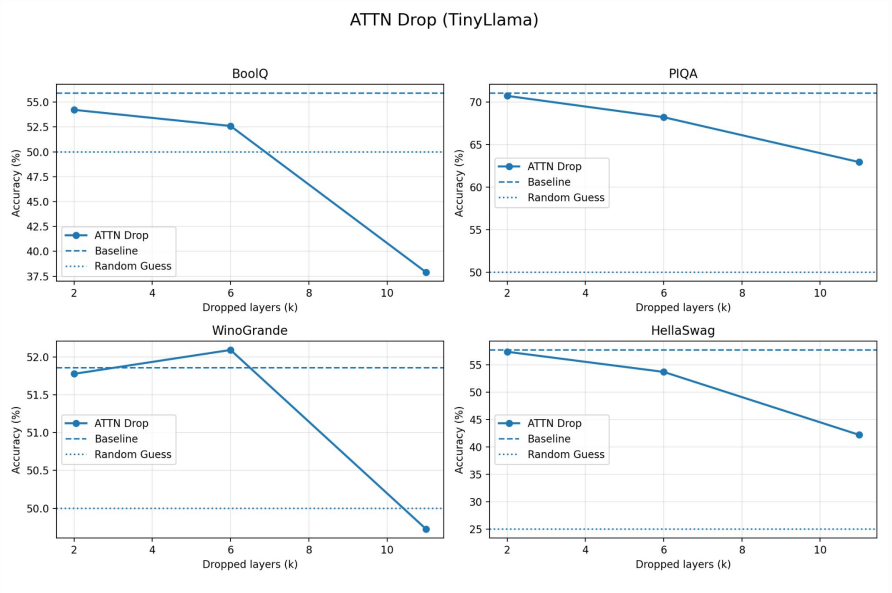
Attention-Drop Results

Removing up to 25% of attention layers has only a small effect on performance.

PIQA and **WinoGrande** remain stable even at 50% pruning, showing that many deeper attention layers are redundant.

In contrast, **BoolQ** and **HellaSwag** degrade more noticeably at higher drop ratios, indicating that they rely more on the full attention stack.

At the language modeling level, perplexity increases moderately (14.6 → 23 at 25% drop), but the model remains usable for most applications.



TinyLlama tolerates attention pruning well, confirming that attention redundancy persists even at smaller model scale.

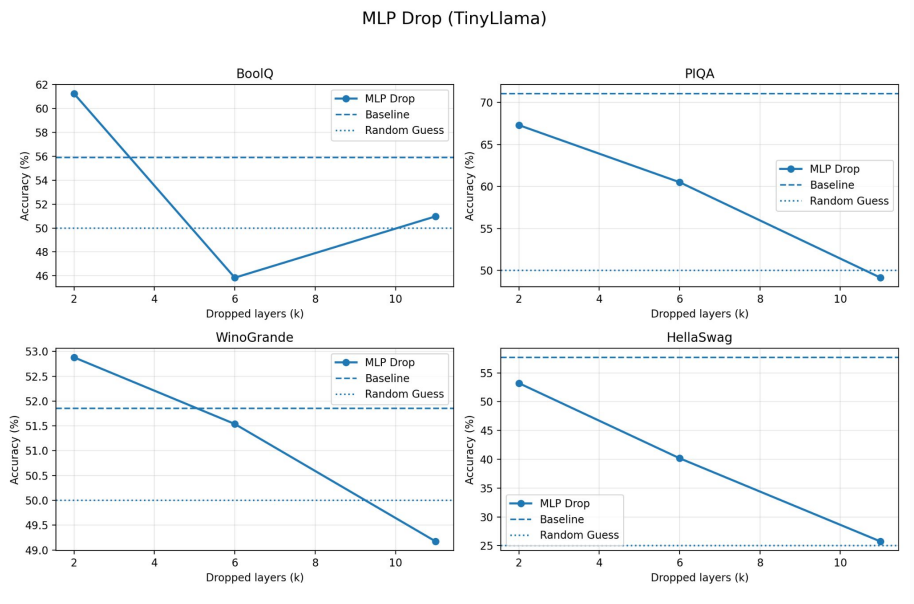
MLP Drop Results

Removing MLP layers leads to a much faster performance decline than attention pruning.

All benchmarks are affected, with the largest drops on **PIQA** (–22 pts) and especially **HellaSwag** (–32 pts at 50% pruning).

BoolQ and **WinoGrande** degrade more gradually, but still show reduced accuracy as more MLP layers are removed.

At the language modeling level, perplexity increases sharply, indicating that MLP pruning quickly harms the model's generative ability.



Even at moderate pruning (25%), MLP Drop performs worse than Attention Drop on average, confirming that MLP layers are less redundant and more important.

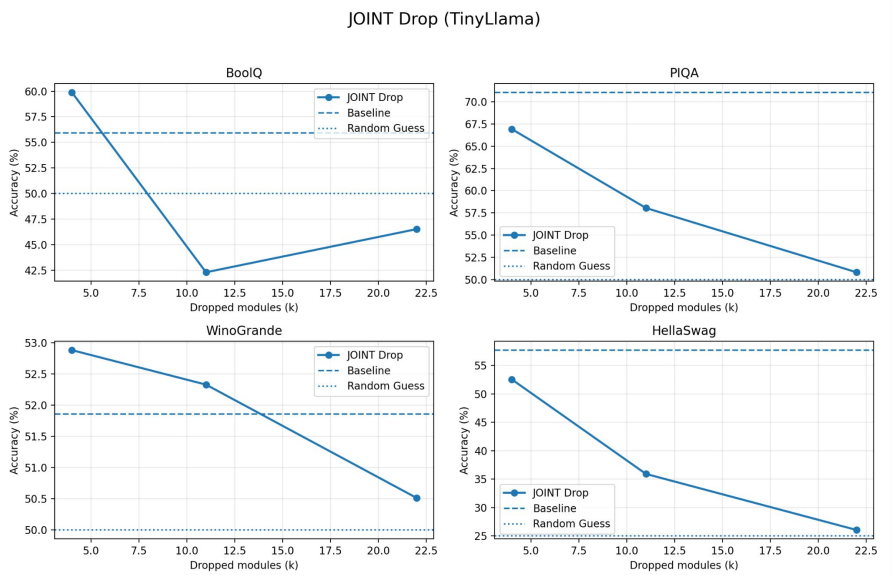
Joint Drop Results

Joint pruning (removing both attention and MLP layers) leads to an earlier and stronger performance collapse.

While **PIQA** and **WinoGrande** remain relatively robust even at high pruning levels, tasks like **BoolQ** and especially **HellaSwag** degrade much more rapidly, showing that they depend on the full Transformer stack.

Compared to single-component pruning, Joint Drop performs the worst at moderate pruning ratios (e.g., 25%), indicating that removing both types of layers compounds the performance loss.

Unlike results reported for larger models, we do not observe any benefit from joint pruning at this smaller scale, suggesting that redundancy patterns change with model size.



Summary & Findings

Attention redundancy generalizes: up to **25% attention layers can be removed** with minimal accuracy loss.

MLP layers are more critical: their importance is **~3× higher** than attention, and pruning them quickly harms performance and perplexity.

Joint pruning is not beneficial: it degrades accuracy earlier and brings no improvement over attention-only pruning.

Method	BoolQ	PIQA	WinoGrande	HellaSwag
Baseline	55.90	71.06	51.85	57.69
Attn-10	54.22	70.73	51.78	57.36
Attn-25	52.60	68.23	52.09	53.68
Attn-50	37.89	62.95	49.72	42.19
MLP-10	61.25	67.30	52.88	53.22
MLP-25	45.84	60.50	51.54	40.18
MLP-50	50.98	49.13	49.17	25.74
Joint-10	59.88	66.92	52.88	52.53
Joint-25	42.29	58.05	52.33	35.92
Joint-50	46.51	50.82	50.51	26.06

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mT5-small

Seq2Seq Translation

mT5 section roadmap

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Task & Setup

Dataset, Model, Metrics

02

Ablation protocol

Dropping Settings

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Encoder results

D0 vs D1 experiments

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Decoder results

D0 vs D1 experiments

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Encoder vs Decoder

Sensitivity Summary

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Key findings

Main Takeaways

Task & Setup (Ro→En Translation)

- **Model:** mT5-small (8 encoder + 8 decoder layers)
- **Task:** Romanian → English (seq2seq translation)
- **Data:** WMT16 Ro–En (20k train / 1k val) + **512 calibration**
- **Training:** 3 epochs, AdamW, lr = 5e–5, max length = 128
- **Decoding & selection:** beam=4, best checkpoint by **SacreBLEU** (baseline **16.90**)

SacreBLEU

- Word/phrase overlap with reference
- Higher is better
- Standardized for fair comparison

chrF++

- Character overlap with reference
- More robust to morphology/spelling
- Higher is better

Beam search (beam = n)

- Keeps the top *n* candidates while decoding
- Expands step-by-step, always keeping the best *n*

Ablation Protocol (Structured Layer Dropping)

What we drop (mT5 components)

- **Encoder:** Self-Attention (SA) blocks **vs** MLP blocks
- **Decoder:** Self-Attention (SA), **Cross-Attention (CA)**, **vs** MLP blocks
- We drop **$k \in \{1,2,3\}$ least-important layers** per component (from ranking).

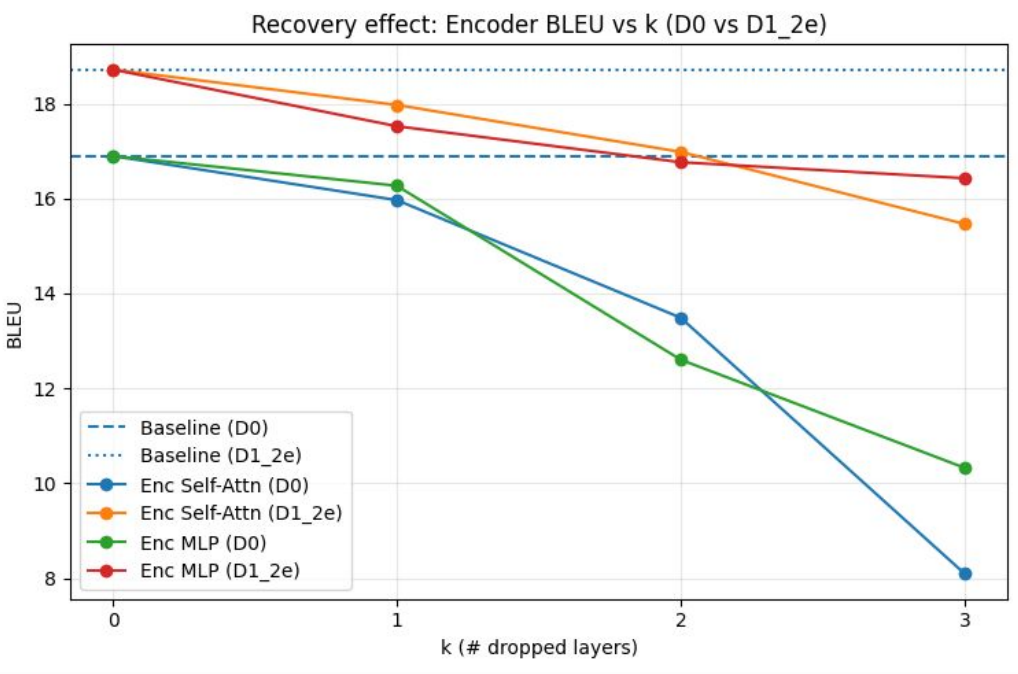
Evaluation settings (D0 vs D1)

- **D0 (no recovery):** drop layers → **evaluate immediately** (no training)
- **D1 (recovery):** drop layers → **fine-tune 1 epoch / 2 epochs** on same train set
  - denoted **D1_1e** and **D1_2e**
- Report **SacreBLEU + chrF++** for all configurations

Goal: **measure redundancy** = how much quality drops when removing “least important” blocks.

Encoder Results (D0 vs D1)

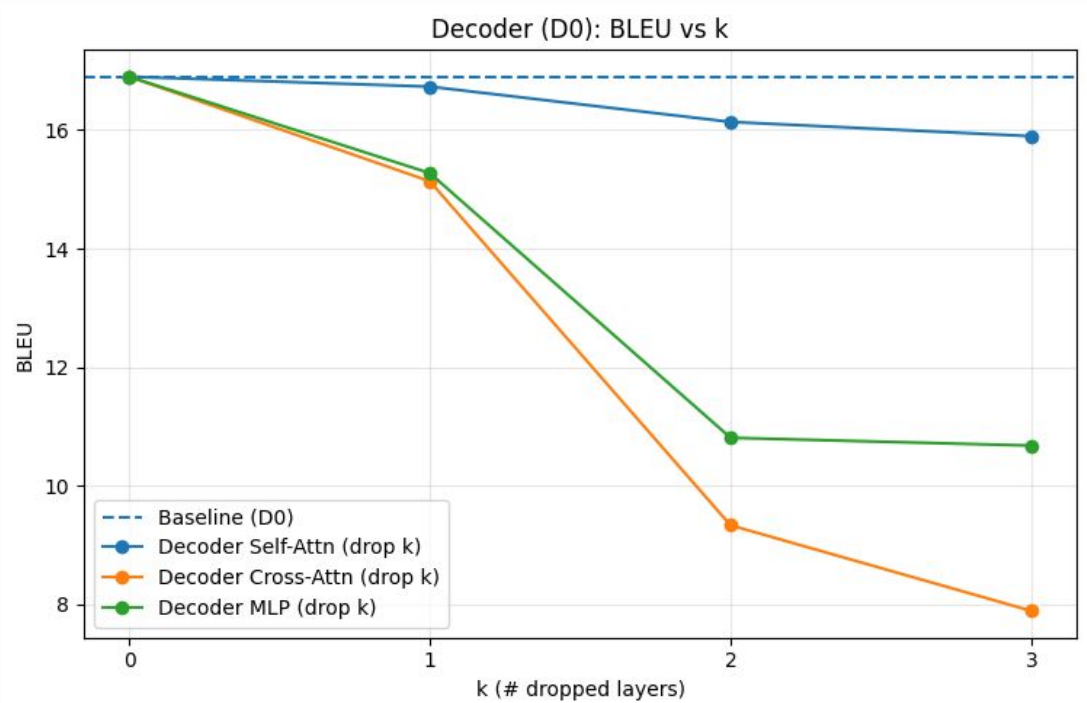
- **D0 (no recovery):** quality drops quickly as **k** increases (worst at **k=3**).
- **D1_2e (recovery):** fine-tuning **recovers most of the loss**.
- **Self-attn vs MLP:** under D0, **MLP is slightly more robust**; after recovery, **both are similar**.
- **Takeaway:** encoder pruning is feasible, but **aggressive dropping needs recovery**.



Metric shown: **SacreBLEU** (chrF++ shows the same trend).

Decoder Results (D0: No Recovery)

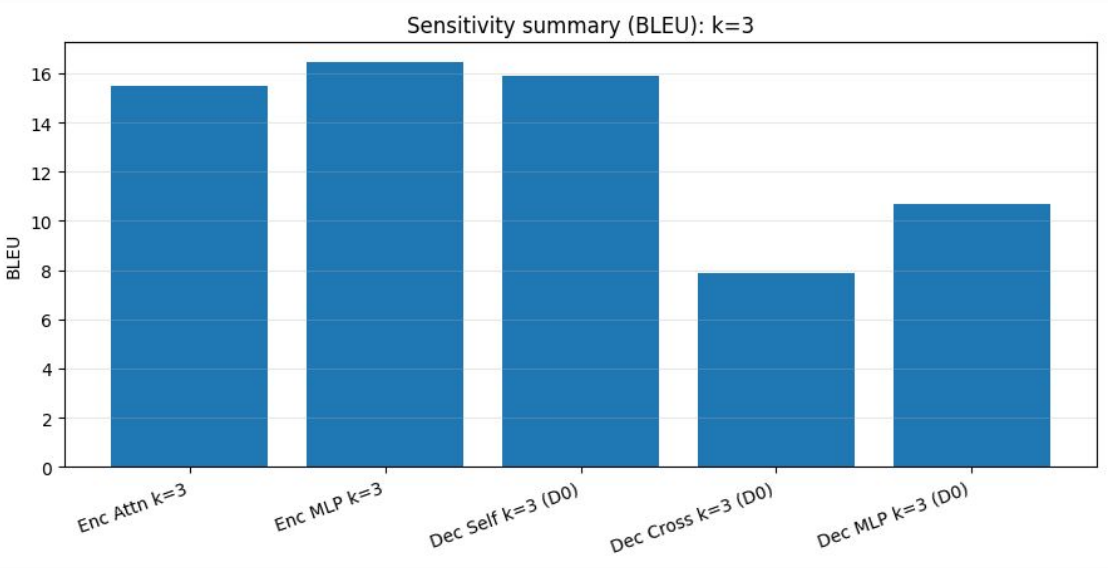
- **Self-attention is robust:** BLEU drops only slightly even at **k = 3** (stays close to baseline).
- **Cross-attention is critical:** performance **collapses fast** as layers are removed (≈16.9 → ≈7.9 at **k = 3**).
- **MLP is intermediate:** noticeable degradation, but **less severe than cross-attention**.
- **Takeaway:** in seq2seq translation, **decoder cross-attention is the least redundant component** and dominates quality loss under aggressive dropping.



Metric shown: **SacreBLEU** (chrF++ shows the same trend).

Encoder vs Decoder (D0, k=3)

- **Encoder is more tolerant:** dropping 3 layers keeps ~15–16 BLEU (SA & MLP).
- **Decoder is asymmetric:**
  - **Self-attn** stays near baseline (~15.9 BLEU)
  - **MLP** drops more (~10.7 BLEU)
  - **Cross-attn** collapses (~7.9 BLEU)
- **Main message: decoder cross-attention is the bottleneck** for seq2seq; pruning it hurts far more than pruning encoder layers.



Metric shown: **SacreBLEU** (chrF++ shows the same trend).

Key Findings (mT5 Ro→En)



Recovery works

1–2 epochs of fine-tuning recovers most of the drop after pruning.

Encoder redundancy exists

Under D0, dropping encoder SA/MLP hurts; with recovery it becomes manageable.

Decoder is asymmetric

Self-attn is robust, MLP is mid-sensitive, cross-attn is highly sensitive.

Bottleneck

Decoder cross-attention is the least redundant component in seq2seq translation.

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Conclusion

Key Takeaways

Attention layers are more redundant than MLP layers

- Observed consistently in RoLLaMA2-7B and TinyLlama-1.1B
- Performance degrades gradually under attention pruning

Redundancy depends on Transformer component

- Encoder layers recover well after fine-tuning
- Decoder cross-attention is critical for translation

Pruning must be component-aware

- Decoder-only vs encoder–decoder models behave differently
- One-size-fits-all pruning is unsafe